Kai Cheng

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Research Interests

Robot Learning, Dexterous Manipulation, Hand-Object Interaction, 3D Vision, Representation Learning

Education

Zhejiang University — Aug 2024 – Jun 2028 (Expected)
B.Eng. in Mechanical Engineering, School of Mechanical Engineering; GPA: 3.57/4.00 — *Hangzhou, China*
Coursework: Machine Learning, Data Structures & Algorithms, Automatic Control, Mechanisms & Machine Theory

Research Experience

Research Intern — May 2026 – Present
X-Dimensional Representations Lab, Zhejiang University, Supervised by Prof. Yiyi Liao — *Hangzhou, China*

- Exploited low-freq RoPE to embed vision-info for dexterous robot policy learning in OpenPI/$\pi_{0.5}$ VLA action tokens.
- Implemented geometry-aware camera pose injection in the OpenPI/$\pi_{0.5}$ LoRA training stack.

Research Intern — Mar 2026 – Present
State Key Laboratory of CAD&CG, Zhejiang University, Supervised by Prof. Hao Chen — *Hangzhou, China*

- Building an end-to-end data pipeline from HOI videos to simulation-ready dexterous manipulation data for RL.
- Developing a feedforward HOI model for high-throughput 3D reconstruction of interacting hands and objects.

Research Intern — Sep 2025 – Feb 2026
State Key Laboratory of CAD&CG, Zhejiang University, Supervised by Prof. Zhaopeng Cui — *Hangzhou, China*

- Worked on learning-based local planning for mobile robots, focusing on safe trajectory generation and robust navigation.
- Built training and evaluation pipelines and visualized planned trajectories in simulated navigation environments.

Selected Projects

Awesome Dexterous Hands — Jun 2026 – Present

- Created and maintain a nine-category open-source research map structured around the human-to-robot dexterous manipulation pipeline.

Stereo HOI Object Pose Estimation — May 2026 – Present

- Built a stereo-video pipeline for temporally consistent 6D object pose tracking in hand-object interaction sequences, integrating stereo depth, multi-view fusion, and temporal smoothing.

Awards & Training

Mathematical Contest in Modeling (MCM/ICM) – Honorable Mention — May 2026
COMAP, Team Leader — International

- Led the team in modeling battery discharge behavior and coordinating model validation and technical reporting.

AI4Math Summer School: Lean 4 and Mathematical Formalization — Jul 2025
Shanghai Jiao Tong University (SJTU) — Shanghai, China

- Awarded “Excellent Participant” for outstanding performance in coursework and assessment.
- Completed hands-on exercises in mathematical formalization using the Lean 4 theorem prover.

Technical Skills

Programming	Python, C++, MATLAB, LaTeX
Machine Learning / CV	PyTorch, OpenCV, NumPy, scikit-learn
Robotics / 3D Vision	ROS 2, MuJoCo, Isaac Sim, stereo vision, camera calibration, motion planning
Mechanical / Hardware	SolidWorks, CAD modeling, mechanical design, 3D printing
Tools	Git, Linux, Lean 4